Tech Stack

Area	Choice	Notes
Engine	Godot 4.7 stable	Installed at `/Users/sumit/Downloads/Godot.app`
Language	GDScript, static typing	No C#: it adds .NET to the APK and complicates Android export
Renderer	Compatibility (OpenGL ES 3.0)	Widest Android support, best for 2D. Mobile (Vulkan) not needed
Physics	Godot Physics 2D	60 Hz, 24 solver iterations, blocks start asleep. Stability test passes (drift < 2 px over 10 s awake), so Rapier is not needed for now
Resolution	1280×720 base, `canvas_items` stretch, `expand` aspect	20:9 phones see 1600×720; HUD anchors to screen edges
Data	JSON in `data/`	Levels, materials, projectiles, balance, store catalogue
Save	`user://save.json` via `Save` autoload	Versioned, written atomically (temp file + rename)
Ads	Google AdMob via Poing Studios `godot-admob-plugin` (Android)	Includes UMP consent. Check the release that matches Godot 4.7 when integrating
IAP	Official GodotGooglePlayBilling 3.3.0 (`addons/GodotGooglePlayBilling`)	Billing Library 9.1
Analytics / crashes	Firebase Analytics + Crashlytics (+NDK) via our plugin CastleFirebase (`addons/castle_firebase`, source `tools/android_plugins/firebase`)	Off until `google-services.json` is in the project root
Push notifications	Firebase Cloud Messaging via **GodotX Firebase 3.1.0** (Core + Messaging, MIT, `addons/godotx_firebase`, AARs in `android/firebase_core`, `android/firebase_messaging`)	Same switch as above; topics in `data/notifications.json`
Play review + updates	CastlePlay class in the same plugin (`com.google.android.play:review:2.0.2`, `app-update:2.1.0`)	Code only, no console setup
Play Games sign-in	CastlePlay, `com.google.android.gms:play-services-games-v2:21.0.0`; project id 300021741882 in `notify/src/main/res/values/games_ids.xml`	Automatic sign-in at start, Settings → Play Games → Sign in. No cloud save yet
Notifications	Local notifications via our plugin CastleNotify (`addons/castle_notify`, source `tools/android_plugins/notify`)	AlarmManager, no server
Intro video	Ogg Theora + WAV in `assets/intro`, captured from the studio HTML by `tools/capture_intro.py` (Playwright + Chrome + imageio-ffmpeg)	

Area	Choice	Notes
Build	Gradle build, Android App Bundle (`.aab`) for Play, APK for local testing	Gradle build is required for plugins
Android	min SDK 24, target SDK 36, arm64-v8a + armeabi-v7a	Java 17, SDK at `~/Library/Android/sdk`, build-tools 36.1
Fonts	Lilita One (titles), Fredoka (body)	SIL OFL, `assets/fonts/`
Art tools	Hand-written SVG (Godot imports natively), AI images from the owner's image tool, ImageMagick for cutouts/resizing	
Audio tools	Python + numpy generators in `tools/`, CC0 libraries	`.ogg` for music, `.wav` for short SFX
Android plugin build	`tools/android_plugins` Gradle project (AGP 8.6.1, Kotlin 2.1.21)	`cd tools/android_plugins && ./gradlew assembleRelease`, then copy the AARs into `addons/castle_*/bin/`
Testing	Headless Godot test scripts in `tests/`, USB phone via `adb`	

Versions on the dev machine (2026-10-29)

Godot 4.7.stable.official.5b4e0cb0f · export templates 4.7.stable · Java 17.0.3 · Android build-tools 34–36.1, platforms 29–36.1, NDK 28.2 · Python 3 + numpy 2.5 · ImageMagick 7.